

Supplementary Material for “Interval Optimization Based PV Hosting Capacity Evaluation in Flexible Distribution Networks Considering Uncertainty and Precise Losses”

NOMENCLATURE

A. Indices and Sets

$i \in \mathcal{N}$	Indices and sets for buses.
$l \in \mathcal{L}$	Indices and sets for branches.
$m \in \mathcal{M}$	Indices and sets for SOP ports.
$t \in \mathcal{T}$	Indices and sets for time periods.
$\mathcal{N}_{PV}$	Candidate AC buses sets for PV installation.
DC	Indices for DC link of the SOP.

B. Parameters

$r_l \setminus x_l$	Resistance and reactance of branch l .
$M_{i,l}^r \setminus M_{i,l}^s$	Incidence matrix associating branch l with its sending and receiving buses i .
$I_l^{\max}$	Branch current limit.
$U_{\max} \setminus U_{\min}$	Upper and lower voltage limits for bus i .
$\chi_t^{PV} \setminus \chi_t^l$	Normalized PV output and load profiles.
$P_i^{\ln}$	Maximum load at bus i .
V_n^{DC}	DC link voltage.
$b_R \setminus b_V \setminus b_P$	Ohmic loss coefficient, constant voltage loss coefficient, and no-load loss coefficient for SOPs.
S_m^{MTn}	Rated capacity of the converter at port m .
Δt	Unit time duration.
$e_{\min} \setminus e_{\max}$	Minimum and maximum state of charge for ES.
$E_{ESn} \setminus P_{ESn}$	Rated capacity and power of ES.
$c_R \setminus c_V \setminus c_P$	Internal resistance loss coefficient, constant voltage loss coefficient, and auxiliary loss coefficient for ESSs.
$S_i^{\max}$	Maximum PV capacity installable at candidate buses due to site area and policy constraints.
$\chi_{i,t}^{PVl} \setminus \chi_{i,t}^{PVu}$	Lower and upper bounds of PV output fluctuation interval.
$\chi_{i,t}^{ll} \setminus \chi_{i,t}^{lu}$	Lower and upper bounds of load fluctuation interval.
$\mathbf{R}_T$	T -by- T temporal covariance matrix.
$\mathbf{R}_S$	N_{PV} -by- N_{PV} spatial covariance matrix.
$\mathbf{R}_T^{-1} \setminus \mathbf{R}_S^{-1}$	Inverse of temporal and spatial covariance matrices $\mathbf{R}_T \setminus \mathbf{R}_S$.
$\Gamma^{TC} \setminus \Gamma^{SC}$	Robust budget of ellipsoidal uncertainty set.
$\mathbf{L}_1 \setminus \mathbf{L}_2$	Lower triangular matrices from Cholesky decomposition, $\mathbf{R}_T^{-1} = \mathbf{L}_1 \mathbf{L}_1^T$, $\mathbf{R}_S^{-1} = \mathbf{L}_2 \mathbf{L}_2^T$.
$N_{PV}^{\min} \setminus N_{PV}^{\max}$	Minimum and maximum number of PV installations.

$\mathbf{P}^{\ln} \setminus \mathbf{S}^{PV \max}$

$\hat{P}_{l,t}^k \setminus \hat{Q}_{l,t}^k \setminus \hat{\beta}_{l,t}^k \setminus \hat{u}_{i,t}^k$	Parameter vectors for maximum load and PV capacity at buses.
$\hat{I}_{m,t}^{DC,k}$	Active power, reactive power, current, and voltage squared at operating point k .
γ^{pu}	SOP port current at operating point k .
$N \setminus \Delta \setminus M_b$	Penalty coefficient for convexification error.
$\mathbf{b}_{eq} \setminus \mathbf{b}_{ineq} \setminus \mathbf{E}_{eq} \setminus \mathbf{J}_{eq} \setminus \mathbf{W}_{ineq} \setminus \mathbf{V}_{ineq}$	Number of segments, interval length, and big-M parameter in binary expansion method.
$\mathbf{c}$	Parameter matrices for linear equality and inequality constraints.
γ^{pu}	Objective function parameters.
$\gamma_{\max}^{pu}$	Penalty coefficient.
	Maximum penalty coefficient.

C. Decision Variables

$S_i^{PV} \setminus \mathbf{S}^{PV}$	Installed capacity of PV and its variable matrix.
$\beta_{l,t} \setminus u_{i,t}$	Square of branch current and bus voltage magnitude.
$P_{i,t}^{PV} \setminus \mathbf{P}_t^{PV}$	Generated power of PV and its variable matrix.
$P_{i,t}^l \setminus \mathbf{P}_t^l$	Active load at bus i and its variable matrix.
$Q_{i,t}^l$	Reactive power of load at bus i .
$P_{i,t}^{MT} \setminus Q_{i,t}^{MT}$	Active and reactive power injected by SOP into AC bus.
$P_{l,t} \setminus Q_{l,t}$	Active and reactive power flow in branch l .
$I_{m,t}^{DC}$	Current flowing from the DC bus to the m -th port.
I_t^{ES}	Magnitude of the current output from the DC energy storage system.
$I_{i=DC,t}^{PV}$	Magnitude of current injected by PV into the DC link.
P_t^{ES}	Active power output from ES.
$P_{m,t}^{MT,loss}$	Loss at the m -th port of SOP.
E_t^{ES}	State of energy of the ideal battery.
$P_t^{ES,ideal}$	Charging and discharging power of the ideal battery.
$P_t^{ES,loss}$	Losses of ES.
$z_i^{PV} \setminus \mathbf{z}^{PV}$	PV placement variable and its variable matrix.
$\tilde{\chi}_{i,t}^{PV} \setminus \tilde{\chi}_{i,t}^l$	PV output and load uncertainty variables, normalized values.

$\tilde{\chi}_i^{\text{PV}} \setminus \tilde{\chi}_t^{\text{PV}}$	Column vectors of normalized PV uncertainty output in temporal and spatial dimensions, $\tilde{\chi}_i^{\text{PV}} = \{\tilde{\chi}_{i,1}^{\text{PV}}, \dots, \tilde{\chi}_{i,T}^{\text{PV}}\}$, $\tilde{\chi}_t^{\text{PV}} = \{\tilde{\chi}_{1,t}^{\text{PV}}, \dots, \tilde{\chi}_{N_{\text{PV}},t}^{\text{PV}}\}$.
$\tilde{\mathbf{a}}$	Binary $(N_{\text{PV}} + 1)$ -by-1 variable matrix indicating whether PV is installed at candidate buses.
$\mathbf{u}$	Uncertainty variable matrix.
$\mathbf{x}$	Linear variables of the inner problem.
$\mathbf{s}$	Second-order cone variables of the inner problem.
$s_{l,t,k}^{\text{PF}} \setminus s_{m,t,k}^{\text{DC}}$	Slack variables for power flow and SOP loss constraints.
$F_{\text{pu}} \setminus F_{\text{PF}} \setminus F_{\text{DC}}$	Penalty terms associated with DCC.
$\setminus F_{\text{ES}}$	
$\xi_{l,t,k}$	Intermediate variables introduced for convexifying BFM.
$\kappa_{1,m,t} \setminus \kappa_{2,m,t} \setminus \kappa_{3,t} \setminus \kappa_{4,t}$	Intermediate variables introduced for convexifying the MSOP-ES model.
$z_n^a \setminus v_n^a$	Auxiliary variables in binary expansion method, including binary and continuous variables.
$\lambda \setminus \mu \setminus \pi_t^p \setminus \pi_t^l$	Dual multipliers.
$\setminus \pi^s$	

D. Functions and Feasible Regions

$\mathcal{U}_{\text{PV}} \setminus \mathcal{U}_{\text{load}} \setminus \mathcal{U}_{\text{siting}}$	Uncertainty sets representing PV output uncertainty, load uncertainty, and PV placement uncertainty.
$\mathbb{E}(\bullet)$	Arithmetic mean.
$\Lambda(\bullet)$	Function converting a column vector to a diagonal matrix.
$\mathcal{K}^{(*)}$	Second-order cone domain with the superscript denoting its dimension.

SUPPLEMENTARY MATERIAL FOR THE MANUSCRIPT

Due to page limitations, we have to place some detailed simulation parameter settings from the manuscript into the supplementary material. In subsequent revisions, the following portions may be appropriately added to the manuscript for the convenience of reviewers and readers.

A. Modified IEEE 33-bus Test Systems

The FDN is formed by interconnecting four modified IEEE 33-bus test systems through SOPs, as shown in Fig. 3 [28]. The nominal voltage is 12.66 kV, and the power base is set at 10 MW. The maximum load is 14.86 MW and 9.2 Mvar. The permissible voltage range for buses is 0.95–1.05 p.u., and the maximum allowable branch current is 450 A (equivalent to 0.5697 p.u.). The four SOP ports are connected to bus 22 of Network 1, bus 25 of Network 2, bus 33 of Network 3, and bus 15 of Network 4, respectively. The rated capacity of each AC/DC converter is 1,000 kVA. A 2,000 kW/4,000 kWh DC energy storage system is connected to the SOP DC link. By fitting simulation data [16], the coefficients of the SOP loss polynomial (b_R , b_V , b_P) are determined as 0.019, 0.005, and 0, respectively. Similarly, the coefficients of the ES loss polynomial (c_R , c_V , c_P) are determined as 0, 0.010, and 0, respectively. The maximum allowable PV installation capacity at each candidate bus is set at 3 MW. For the iterative solution algorithm, the maximum number of iterations $MaxIter$ is set to 100, the initial penalty coefficient γ^{pu} is set to $1e-2$, and the maximum penalty coefficient $\gamma_{\text{max}}^{\text{pu}}$ is set to 2.